

THE DISTRIBUTION OF LINEAR AND QUADRATIC FORMS IN COMPLEX NORMAL VECTORS

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Key words: Conditional distribution, Second degree polynomial, Moment generating function, Moments, Conditions for Wishartness

1. Introduction and Summary

The complex normal distribution has been studied by Wooding (1956), Turin (1960) and Goodman (1963) and distribution problems concerning complex normal variables have been dealt with by James (1964), Khatri (1965), De Waal (1968) and others. There are examples in which complex measurements are involved (see e.g. Turin (1960) for examples in physics).

This paper deals with (i) certain conditional distributions of complex normal variables and (ii) the distribution of second degree polynomials in complex normal vectors.

The conditional distribution of a complex normal vector, given certain linear combinations among its elements, is derived.

The moment generating function of a general non-central polynomial of the second degree in complex normal vectors is considered. The j -th cumulant of the trace of such a polynomial is also given. With these results it is possible to establish necessary and sufficient conditions that a polynomial of the second degree in complex normal vectors has a non-central complex Wishart distribution; these conditions are given in the last section.

2. Notation

- (i) Lower-case letters will denote scalars, the vector $(x_1, x_2, \dots, x_n)$ will be denoted by $\underline{x}'$ and capital letters will denote matrices.
- (ii) The matrix X' is the transpose of X , $\overline{X}$ is the conjugate of X and $\overline{X}'$ is the hermitian transpose of X .
- (iii) If the real matrix X is positive definite, then $X^{1/2}$ is the symmetric square root of X . If the complex matrix X is positive definite, then $X^{1/2}$ is the hermitian square root of X .
- (iv) The trace of X will be denoted by $\text{tr } X$.

(v) $\exp(\text{tr } X)$ will be denoted by $\text{etr } X$.

3. The complex normal distribution

Let $x = x_R + ix_I$ be a complex random variable, whose real and imaginary parts, x_R and x_I , are independently normally distributed with the same variance, $\frac{1}{2}\sigma^2$, $E(x_R) = \mu_R$ and $E(x_I) = \mu_I$. Let $\mu = \mu_R + i\mu_I$. The joint distribution of x_R and x_I is called the complex normal distribution, $CN(\mu, \sigma^2)$, with density function given by:

$$(3.1) \quad f(x) = (\pi\sigma^2)^{-1} \exp\{-\sigma^{-2}(x_R - \mu_R)^2 - \sigma^{-2}(x_I - \mu_I)^2\} \\ = (\pi\sigma^2)^{-1} \exp\{-\sigma^{-2}(x - \mu)(\overline{x - \mu})\}$$

The complex variance of x is defined by:

$$\text{var}(x) = E\{(x - \mu)(\overline{x - \mu})\} = \sigma^2$$

and $E(x) = \mu_R + i\mu_I = \mu$.

Suppose $\underline{x} = \underline{x}_R + i\underline{x}_I$ is a $p \times 1$ complex random vector, whose real and imaginary parts, $\underline{x}_R$ and $\underline{x}_I$, have a joint $2p$ -variate normal distribution with covariance matrix of the form:

$$E\{(\underline{x}_0 - \underline{\mu}_0)(\underline{x}_0 - \underline{\mu}_0)'\} = \frac{1}{2} \begin{bmatrix} \Sigma_R & -\Sigma_I \\ \Sigma_I & \Sigma_R \end{bmatrix} = \Sigma_0, \text{ say,}$$

where $\underline{x}'_0 = (\underline{x}'_R, \underline{x}'_I)$, $\underline{\mu}'_0 = (\underline{\mu}'_R, \underline{\mu}'_I)$, $\underline{\mu}_R = E(\underline{x}_R)$, $\underline{\mu}_I = E(\underline{x}_I)$ and Σ_I is skew symmetric. Let $\Sigma = \Sigma_R + i\Sigma_I$ and $\underline{\mu} = \underline{\mu}_R + i\underline{\mu}_I$. The joint distribution of $\underline{x}_R$ and $\underline{x}_I$ is called the complex multivariate normal distribution and Goodman (1963) proved that the joint density of $\underline{x}_R$ and $\underline{x}_I$ may be written in the form:

$$(3.2) \quad f(\underline{x}) = \pi^{-p} |\Sigma|^{-1} \exp\{-\overline{(\underline{x} - \underline{\mu})}' \Sigma^{-1} (\underline{x} - \underline{\mu})\}$$

The complex covariance matrix of $\underline{x}$ is defined as:

$$\text{Cov}(\underline{x}, \underline{x}') = E\{(\underline{x} - \underline{\mu})(\overline{\underline{x} - \underline{\mu}})'\} = \Sigma, \Sigma \text{ being hermitian.}$$

If $\text{rank } \Sigma_0 = 2r$, then there exists a real matrix

$$A_0 = \begin{bmatrix} A_1 & -B_1 \\ B_1 & A_1 \end{bmatrix}, \quad A_1 : p \times r, B_1 : p \times r, \text{ such that}$$

$\underline{x}_0 = A_0 \underline{z}_0 + \underline{\mu}_0$, where $\text{rank } A_0 = 2r = 2 \text{ rank } \Sigma$, $\underline{z}'_0 = (\underline{z}'_R \ \underline{z}'_I)$, and $\underline{z}_R:rx1$ and $\underline{z}_I:rx1$ are independently $N(\underline{0}, I)$ distributed. Thus:

$$(3.3) \quad \underline{x} = A\underline{z} + \underline{\mu}$$

where $A = A_1 + iB_1$, $\underline{z} = \underline{z}_R + i\underline{z}_I$, $\text{rank } A = r$ and $\underline{z}$ has a $CN(\underline{0}, I)$ distribution. If $r < p$, we say that $\underline{z}$ has a singular complex normal distribution.

4. Conditional Distributions of Complex Normal Variates

Lemma 4.1

(i) $RK\bar{K}' = 0$ if and only if $RK = 0$ and

(ii) $K\bar{K}'B = 0$ if and only if $\bar{K}'B = 0$.

Proof:

If $RK = 0$, then $RK\bar{K}' = 0$. If $RK\bar{K}' = 0$, then $RK(\overline{RK})' = 0$, i.e. $RK = 0$. This proves (i). Part (ii) is proved similarly.

Definition 4.1:

A generalized inverse of the complex matrix $A:mxn$ is defined as a matrix $A^*:nxm$ that satisfies the condition:

$$(4.1) \quad AA^*A = A$$

The existence of a generalized inverse for any complex matrix may be proved similarly to the case of real matrices. Using Lemma 4.1, it follows that:

$$C(i) \quad \bar{A}'(A\bar{A}')^*A\bar{A}' = \bar{A}'$$

$$C(ii) \quad A\bar{A}'(A\bar{A}')^*A = A$$

C(iii) The matrix $\bar{A}'(A\bar{A}')^*A$ is unique, i.e. independent of the choice of a generalized inverse of $A\bar{A}'$, hermitian and idempotent.

C(iv) $\overline{(\bar{A})^*}$ is also a generalized inverse of A .

Theorem 4.1:

Suppose $\underline{x}:nx1$ has a $CN(\underline{\mu}, \Sigma)$ distribution and Σ may be singular. Let $\underline{x}' = (\underline{x}'_1, \underline{x}'_2)$,

where $\underline{x}_1:qx1$, $\underline{x}_2:(n-q)x1$ and $\underline{\mu}' = (\underline{\mu}'_1, \underline{\mu}'_2)$,

$$\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix},$$

correspondingly. The conditional distribution of $\underline{x}_1$, given $\underline{x}_2$, is the complex normal distribution with:

$$(4.2) \quad E(\underline{x}_1/\underline{x}_2) = \underline{\mu}_1 + \Sigma_{12}\Sigma_{22}^*(\underline{x}_2 - \underline{\mu}_2)$$

$$(4.3) \quad \text{Cov}(\underline{x}_1, \underline{x}_1'/\underline{x}_2) = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^*\Sigma_{21}$$

The expressions (4.2) and (4.3) are independent of the choice of a generalized inverse of Σ_{22} , and $\text{rank}(\Sigma_{11} - \Sigma_{12}\Sigma_{22}^*\Sigma_{21}) = \text{rank } \Sigma - \text{rank } \Sigma_{22}$.

Proof:

$$\text{Let } \underline{y} = \begin{bmatrix} \underline{y}_1 \\ \underline{y}_2 \end{bmatrix} = P \begin{bmatrix} \underline{x}_1 \\ \underline{x}_2 \end{bmatrix}, \text{ where } P = \begin{bmatrix} I & -\Sigma_{12}\Sigma_{22}^* \\ 0 & I \end{bmatrix}.$$

$$\text{Thus } E(\underline{y}) = E \begin{bmatrix} \underline{y}_1 \\ \underline{y}_2 \end{bmatrix} = P\underline{\mu} = \begin{bmatrix} \underline{\mu}_1 - \Sigma_{12}\Sigma_{22}^*\underline{\mu}_2 \\ \underline{\mu}_2 \end{bmatrix} \text{ and}$$

$$\text{Cov}(\underline{y}, \underline{y}') = P\Sigma P' = \begin{bmatrix} \Sigma_{11} - \Sigma_{12}\Sigma_{22}^*\Sigma_{21} & 0 \\ 0 & \Sigma_{22} \end{bmatrix},$$

$$\text{since } \Sigma = \begin{bmatrix} M_1 \bar{M}_1' & M_1 \bar{M}_2' \\ M_2 \bar{M}_1' & M_2 \bar{M}_2' \end{bmatrix}, \text{ thus}$$

$$\Sigma_{11} - \Sigma_{12}\Sigma_{22}^*\Sigma_{21} = M_1 \bar{M}_2' - M_1 \bar{M}_2'(M_2 \bar{M}_2')^* M_2 \bar{M}_2'$$

$$= M_1 \bar{M}_2' - M_1 \bar{M}_2'$$

$$= 0$$

and $\Sigma_{21} - \Sigma_{22}\Sigma_{22}^*\Sigma_{21} = 0$, similarly. It follows that $\underline{y}_1$ and $\underline{y}_2$ are independent. But $\underline{x}_1 = \underline{y}_1 + \Sigma_{12}\Sigma_{22}^*\underline{y}_2$ and $\underline{x}_2 = \underline{y}_2$. Thus (4.2) and (4.3) are proved. Using properties C(i), C(ii), C(iii) and C(iv), the invariance of (4.2) and (4.3) with respect to the choice of the generalized inverse of Σ_{22} , could be established analogous to the real case (see Pringle

and Rayner (1971), pp. 71-72).

The expressions (4.2) and (4.3) are also valid if the real vector $\underline{x}$ has a $N(\underline{\mu}, \Sigma)$ distribution. In this case the corresponding results are given by Marsaglia (1964) and Pringle and Rayner (1971).

Theorem 4.2:

Suppose $\underline{x}: n \times 1$ has a $CN(\underline{\mu}, \Sigma)$ distribution and Σ may be singular. Let $\underline{y} = L\underline{x}$, where $L: s \times n$. The conditional distribution of $\underline{x}$, given $\underline{y}$, is the complex normal distribution with:

$$(4.4) \quad E(\underline{x}|\underline{y}) = \underline{\mu} + \overline{(L\Sigma)'}(L\Sigma\bar{L}')^*(\underline{y} - L\underline{\mu}) \\ = \{I - \overline{(L\Sigma)'}(L\Sigma\bar{L}')^*L\}\underline{\mu} + \overline{(L\Sigma)'}(L\Sigma\bar{L}')^*\underline{y}$$

$$(4.5) \quad \text{Cov}(\underline{x}, \underline{x}'|\underline{y}) = \Sigma - \overline{(L\Sigma)'}(L\Sigma\bar{L}')^*L\Sigma$$

The expressions (4.4) and (4.5) are independent of the choice of a generalized inverse of $L\Sigma\bar{L}'$ and $\text{rank}\{\Sigma - \overline{(L\Sigma)'}(L\Sigma\bar{L}')^*L\Sigma\} = \text{rank}\Sigma - \text{rank}(L\Sigma\bar{L}')$.

Proof:

Let $\underline{y} = \begin{bmatrix} I_n \\ L \end{bmatrix} \underline{x}$. Then $\underline{y}: (s+n) \times 1$ has a complex normal distribution with

$$E(\underline{y}) = \begin{bmatrix} \underline{\mu} \\ L\underline{\mu} \end{bmatrix}, \text{ and } \text{Cov}(\underline{y}, \underline{y}') = \begin{bmatrix} \Sigma & \overline{(L\Sigma)'} \\ L\Sigma & L\Sigma\bar{L}' \end{bmatrix}.$$

Let $\underline{y}' = (\underline{y}'_1, \underline{y}'_2)$, where $\underline{y}_1: n \times 1$. It follows from theorem 4.1 that the conditional distribution of $\underline{y}_1$, given $\underline{y}_2$, is the complex normal distribution with:

$$E(\underline{y}_1|\underline{y}_2) = \underline{\mu} + \overline{(L\Sigma)'}(L\Sigma\bar{L}')^*(\underline{y}_2 - L\underline{\mu})$$

$$\text{and } \text{Cov}(\underline{y}_1, \underline{y}_1'|\underline{y}_2) = \Sigma - \overline{(L\Sigma)'}(L\Sigma\bar{L}')^*L\Sigma,$$

and these expressions are independent of the choice of a generalized inverse. Also

$$\text{rank}\{\Sigma - \overline{(L\Sigma)'}(L\Sigma\bar{L}')^*L\Sigma\} = \text{rank}\Sigma - \text{rank}(L\Sigma\bar{L}').$$

Since $\underline{y}_1 = \underline{x}$ and $\underline{y}_2 = \underline{v}$, the theorem is proved.

The expressions (4.4) and (4.5) are also valid if the real vector $\underline{x}$ has a $N(\underline{\mu}, \Sigma)$

Proof:

Let $Y = \Sigma_1^{-1/2} X \Sigma_2^{-1/2}$. Thus the columns of Y are independently and complex normally distributed with covariance matrix I and $E(Y) = \Sigma_1^{-1/2} \mu \Sigma_2^{-1/2} = \nu$, say. Thus:

$$S = D_0 Y A_0 \bar{Y}' \bar{D}_0' + \frac{1}{2} (L_{10} \bar{Y}' \bar{L}_{20}' + L_{20} Y \bar{L}_{10}') + C$$

where $D_0 = D \Sigma_1^{1/2}$, $A_0 = \Sigma_2^{1/2} A \Sigma_2^{1/2}$, $L_{10} = L_1 \Sigma_2^{1/2}$ and $L_{20} = L_2 \Sigma_1^{1/2}$. The m.g.f. of S is given by:

$$\begin{aligned} M_S(\theta) &= E[\text{etr}(\theta S)] \\ &= E[\text{etr}\{\bar{D}_0' \theta D_0 Y A_0 \bar{Y}' + \frac{1}{2} (\bar{L}_{20}' \theta L_{10} \bar{Y}' + Y \bar{L}_{10}' \theta L_{20})\} \cdot \text{etr}(\theta C)] \end{aligned}$$

There exists a unitary matrix U , such that:

$$U \bar{D}_0' \theta D_0 U' = \text{diag}(d_1, d_2, \dots, d_p) = \Delta_D, \text{ say,}$$

where $d_i, i=1, \dots, p$ are the characteristic roots of $\bar{D}_0' \theta D_0$. Let $Z = UY$. Thus the columns of Z are independently and complex normally distributed with covariance matrix I , and $E(Z) = U\nu = \xi$, say. Thus:

$$\begin{aligned} M_S(\theta) &= E[\text{etr}\{\Delta_D Z A_0 \bar{Z}' + \frac{1}{2} (U \bar{L}_{20}' \theta L_{10} \bar{Z}' + Z \bar{L}_{10}' \theta L_{20} U')\} \cdot \text{etr}(\theta C)] \\ &= E[\exp\{\sum_{i=1}^p \{d_i z_i' A_0 \bar{z}_i + \frac{1}{2} (\xi_i' \bar{z}_i + z_i \bar{\xi}_i)\} + \text{tr}\{\theta C\}\}] \end{aligned}$$

where z_i' is the i -th row of Z and ξ_i' is the i -th row of $U \bar{L}_{20}' \theta L_{10}$. Since $z_i, i=1, \dots, p$ are independent, it follows from theorem 5.2 that:

$$\begin{aligned} M_S(\theta) &= \prod_{i=1}^p [\exp\{\sum_{k=1}^{\infty} k^{-1} \text{tr}\{(d_i A_0)^k\} - \xi_i' \bar{\xi}_i \\ &\quad + \sum_{k=0}^{\infty} (\xi_i + \frac{1}{2} \bar{\xi}_i)' (A_0 d_i)^k \overline{(\xi_i + \frac{1}{2} \bar{\xi}_i)}\}] \text{etr}(\theta C) \end{aligned}$$

where ξ_i' is the i -th row of ξ . This leads to the expression (5.4).

Corollary 5.1:

If $D=I_p$, $A=\Sigma_2=I_n$ and L_1, L_2 and C are zero matrices, then $S = X \bar{X}'$ has a non-central complex Wishart distribution ($CW(\Sigma, n, \Sigma^{-1} \mu \bar{\mu}')$) and the m.g.f. of S is given by ($\Sigma_1 = \Sigma$):

$$(5.5) \quad M_S(\theta) = \exp\left\{\sum_{k=1}^{\infty} k^{-1} n \text{tr}\{(\Sigma \theta)^k\} - \text{tr}\{\Sigma^{-1} \mu \bar{\mu}'\} + \sum_{k=0}^{\infty} \text{tr}\{\Sigma^{-1} \mu \bar{\mu}' (\theta \Sigma)^k\}\right\}$$

$$= |I - \Sigma\theta|^{-n} \text{etr}\{\theta(I - \Sigma\theta)^{-1} \mu \bar{\mu}'\}$$

The m.g.f. of $\text{tr } S$ follows from theorem 5.3 by putting θ equal to tI :

Theorem 5.4:

Let S be defined as in theorem 5.3. The m.g.f. of $\text{tr } S$ is given by:

$$\begin{aligned} (5.6) \quad M_{\text{tr } S}(t) = \exp \{ & \sum_{k=1}^{\infty} k^{-1} t^k \text{tr}[(\bar{D}' D \Sigma_1)^k] \text{tr}[(A \Sigma_2)^k] - \text{tr}(\Sigma_1^{-1} \mu \Sigma_2^{-1} \bar{\mu}') \\ & + t \text{tr}(C) + \sum_{k=0}^{\infty} \left(t^k \text{tr}[(\bar{D}' D \Sigma_1)^k \Sigma_1^{-1} \mu (A \Sigma_2)^k \Sigma_2^{-1} \bar{\mu}'] \right. \\ & + \frac{1}{2} t^{k+1} \text{tr}[(\bar{D}' D \Sigma_1)^k \bar{L}'_2 L_1 (\Sigma_2 A)^k \bar{\mu}'] \\ & + \frac{1}{2} t^{k+1} \text{tr}[\mu (A \Sigma_2)^k \bar{L}'_1 L_2 (\Sigma_1 \bar{D}' D)^k] \\ & \left. + \frac{1}{4} t^{k+1} \text{tr}[\Sigma_1 (\bar{D}' D \Sigma_1)^k \bar{L}'_2 L_1 \Sigma_2 (A \Sigma_2)^k \bar{L}'_1 L_2] \right) \} \end{aligned}$$

From theorem 5.4 the following result is obtained:

Theorem 5.5:

Let S be defined as in theorem 5.3. The j -th cumulant of $\text{tr } S$ is given by:

$$\begin{aligned} (5.7) \quad \kappa_j = (j-1)! \text{tr}[(\bar{D}' D \Sigma_1)^j] \text{tr}[(A \Sigma_2)^j] + \delta_{1j} \text{tr}(C) \\ + j! \text{tr}[(\bar{D}' D \Sigma_1)^j \Sigma_1^{-1} \mu (A \Sigma_2)^j \Sigma_2^{-1} \bar{\mu}'] \\ + \frac{1}{2} j! \text{tr}[(\bar{D}' D \Sigma_1)^j \bar{L}'_2 L_1 (\Sigma_2 A)^j \bar{\mu}'] \\ + \frac{1}{2} j! \text{tr}[\mu (A \Sigma_2)^j \bar{L}'_1 L_2 (\Sigma_1 \bar{D}' D)^j] \\ + \frac{1}{4} j! \text{tr}[\Sigma_1 (\bar{D}' D \Sigma_1)^j \bar{L}'_2 L_1 \Sigma_2 (A \Sigma_2)^j \bar{L}'_1 L_2] \end{aligned}$$

where $\delta_{1j} = 1$ if $j=1$ and 0 otherwise and $B^j \equiv 0$ for $j < 0$.

[In the real case, i.e. if the density of X : $p \times n$ is:

$$(2\pi)^{-\frac{1}{2}np} |\Sigma_1|^{-\frac{1}{2}n} |\Sigma_2|^{-\frac{1}{2}n} p \text{etr} \{ -\frac{1}{2} \Sigma_1^{-1} (X - \mu) \Sigma_2^{-1} (X - \mu)' \}$$

then the j -th cumulant of $\text{tr } S = \text{tr} \{DXAX'D' + \frac{1}{2}(L_1X'L_2' + L_2XL_1') + C\}$, where all matrices are real, is given by:

$$(5.8) \quad \kappa_j = 2^{j-1}(j-1)! \left[\text{tr}[(D'D\Sigma_1)^j] \text{tr}[(A\Sigma_2)^j] + \delta_{1j} \text{tr}(C) \right. \\ \left. + 2^{j-1}j! \text{tr}[(D'D\Sigma_1)^j \Sigma_1^{-1} \mu(A\Sigma_2)^j \Sigma_2^{-1} \mu'] \right. \\ \left. + 2^{j-1}j! \text{tr}[(D'D\Sigma_1)^{j-1} L_2' L_1 (\Sigma_2 A)^{j-1} \mu'] \right. \\ \left. + 2^{j-3}j! \text{tr}[\Sigma_1 (D'D\Sigma_1)^{j-2} L_2' L_1 \Sigma_2 (A\Sigma_2)^{j-2} L_1' L_2] \right]$$

Corollary 5.2:

Let S be defined as in corollary 5.1. The j -th cumulant of $\text{tr } S$ is given by:

$$(5.9) \quad \kappa_j = (j-1)! n \text{tr}(\Sigma^j) + j! \text{tr}(\Sigma^{j-1} \mu \bar{\mu}')$$

6. Necessary and sufficient conditions that a polynomial of the second degree in complex normal vectors has a non-central complex Wishart distribution.

Suppose $\underline{x}:n \times 1$ has a $CN(\underline{\mu}, I)$ distribution. The m.g.f. of $s = \underline{x}' \underline{x}$ follows from (5.2):

$$(6.1) \quad M_s(\theta) = (1-\theta)^{-n} \exp\{\theta \delta^2 (1-\theta)^{-1}\}$$

where $\delta^2 = \underline{\mu}' \underline{\mu}$. Thus s has a non-central complex χ^2 distribution, $C\chi'^2$ say, with n degrees of freedom and non-centrality parameter δ^2 . It is clear that if s has a $C\chi'^2$ distribution with n degrees of freedom and non-centrality parameter δ^2 , then $2s$ has a χ'^2 distribution with $2n$ degrees of freedom and non-centrality parameter $2\delta^2$.

Theorem 6.1:

If $\underline{x}:n \times 1$ has a $CN(\underline{\mu}, I)$ distribution, then a set of necessary and sufficient conditions that $s = \underline{x}' A \underline{x} + \frac{1}{2}(\underline{\ell}' \underline{x} + \underline{x}' \underline{\ell}) + c$, where A is a hermitian matrix and c a real constant, is $C\chi'^2$ distributed, is given by:

$$(6.2) \quad (i) \quad A^2 = A$$

$$(ii) \quad \underline{\ell}' A = \underline{\ell}'$$

$$(iii) \quad c = \frac{1}{2} \underline{\ell}' \underline{\ell}$$

The number of degrees of freedom is $r = \text{rank } A = \text{tr } A$ and the non-centrality parameter is:

$$\delta^2 = \underline{\mu}' A \underline{\mu} + \frac{1}{2} (\underline{\ell}' \underline{\mu} + \underline{\mu}' \underline{\ell}) + c = (\underline{\mu}' A + \frac{1}{2} \underline{\ell}') \overline{(\underline{A}' \underline{\mu} + \frac{1}{2} \underline{\ell})}$$

Proof:

By equating the m.g.f. of s , given by (5.2), to the m.g.f. of a $C\chi'^2$ distribution with r degrees of freedom, following the same procedure as given by Khatri (1962), lemma 3, the conditions (6.2) follow

Theorem 6.2:

Suppose that the rows of $X:pxn$ are independently and complex normally distributed with covariance matrix I and $E(X) = \underline{\mu}$. A set of necessary and sufficient conditions that $S = XA\bar{X}' + \frac{1}{2}(L\bar{X}' + X\bar{L}') + C$, where A and C are hermitian matrices, has a complex non-central Wishart distribution, is given by:

$$(6.3) \text{ (i) } A^2 = A$$

$$(ii) \quad L = LA$$

$$(iii) \quad C = \frac{1}{2} L\bar{L}'$$

The number of degrees of freedom is $r = \text{tr } A$ and the matrix of non-centrality parameters is $\theta = (\underline{\mu}A + \frac{1}{2}L)/(\underline{\mu}A + \frac{1}{2}L)'$.

Proof:

In particular the i -th diagonal element of S should have a $C\chi'^2$ distribution. From theorem 6.1 it follows that it is necessary that: (i) $A^2 = A$, (ii) $L = LA$ and (iii) the diagonal elements of $C - \frac{1}{2} L\bar{L}'$ should be zero. Thus:

$$S = (Y + \frac{1}{2} L\bar{U}') \overline{(Y + \frac{1}{2} L\bar{U}')} + C - \frac{1}{2} L A \bar{L}'$$

where $\bar{U}'U = A$, $\bar{U}':n \times r$ is a semi-unitary matrix and the rows of $Y:pxr$ are independently and complex normally distributed with $E(Y) = \nu = \underline{\mu}\bar{U}'$ and covariance matrix I . Thus the constant term $C - \frac{1}{2} L A \bar{L}'$ must be a zero matrix. This proves the necessity of the conditions (6.3), and the sufficiency is immediate. The number of degrees of freedom is $r = \text{tr } A$ and the matrix of non-centrality parameters is:

$$\theta = (\nu + \frac{1}{2} L\bar{U}') \overline{(\nu + \frac{1}{2} L\bar{U}')} = (\underline{\mu}A + \frac{1}{2}L) \overline{(\underline{\mu}A + \frac{1}{2}L)'}'$$

Theorem 6.3:

Suppose that the rows of $X:pxn$ are independently and complex normally distributed with covariance matrix V (positive semi-definite) and $E(X) = \underline{\mu}$. Let S be defined as in theorem 6.2. A set of necessary and sufficient conditions that S has a

$CW(I, r, \theta)$ distribution, is given by:

$$(6.4) \quad (i) \quad \bar{V}A\bar{V}A\bar{V} = \bar{V}A\bar{V}$$

$$(ii) \quad (\mu A + \frac{1}{2}L)\bar{V} = (\mu A + \frac{1}{2}L)\bar{V}A\bar{V}$$

$$(iii) \quad \mu A\bar{\mu}' + \frac{1}{2}(L\bar{\mu}' + \mu\bar{L}') + C = (\mu A + \frac{1}{2}L)\bar{V}(\mu A + \frac{1}{2}L)'$$

The number of degrees of freedom is $r = \text{tr}(A\bar{V})$ and $\theta = (\mu A + \frac{1}{2}L)\bar{V}(\mu A + \frac{1}{2}L)'$.

Proof:

Let $q = \text{rank } V$. There exists a matrix $B: q \times n$ such that $X = YB + \mu$, where the rows of $Y: p \times q$ are independently $CN(0, I)$ distributed, and $V = B'\bar{B}$. Thus:

$$\begin{aligned} S = YB\bar{A}\bar{B}'\bar{Y}' + \frac{1}{2} \{ (2\mu\bar{A}\bar{B}' + L\bar{B}')\bar{Y}' + \overline{Y(2\mu\bar{A}\bar{B}' + L\bar{B}')'} \} \\ + \mu A\bar{\mu}' + \frac{1}{2}(L\bar{\mu}' + \mu\bar{L}') + C \end{aligned}$$

The theorem follows from theorem 6.2 and by using lemma 4.1.

ACKNOWLEDGEMENT

The author wishes to express his gratitude to Prof. D.J. de Waal for his valuable suggestions.

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Manuscript received 12 March 1973; revised 31 July 1973.